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COMPUTER PROCESSES FOR CLUSTERING PROPERTIES INTO NEIGHBORHOODS AND GENERATING NEIGHBORHOOD-SPECIFIC MODELS

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(60)

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(51)

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G06N 20/00

(2019.01)

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G06K 9/6289

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G06Q 50/163

(2013.01)

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None

See application file for complete search history.

(56)

References Cited

U.S. PATENT DOCUMENTS

5,361,201

A

11/1994

Jost et al.

8,655,106

B2

2/2014

Den Herder et al.

10,380,653

B1 *

8/2019

Flint

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G06Q 30/0278

2008/0002883

A1

11/2008

Miles et al.

2013/0103595

A1

4/2013

Berry et al.

2014/0164260

A1

6/2014

Spieckerman

2015/0112874

A1 *

4/2015

Serio

.....

G06Q 30/0278

705/306

(Continued)

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(57)

ABSTRACT

A computer system and associated processes are disclosed for grouping similar real estate properties into contiguous neighborhoods, and for generating neighborhood-specific models capable of estimating property values within their respective neighborhoods. A clustering component uses various sources of property-level data to group properties based on measures of property similarity. For example, the clustering component may use features extracted from property images to identify properties with similar characteristics. As another example, the clustering component may measure property similarity based on how frequently specific properties are designated as comparable in appraisal reports. A model generator uses a machine learning process to determine, for specific neighborhoods, correlations between property attributes and values, and uses these correlations to generate the neighborhood specific models.

21 Claims, 5 Drawing Sheets

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graph LR
    subgraph 20 [Computer System]
        direction TB
        PC[Property Clustering]
        subgraph 22 [Property Clustering]
            FE[Feature Extraction 24]
        end
        ND[(Neighborhood Definitions 28)]
        MG[Model Generator 30]
        DB[(AVMs for Specific Neighborhoods 34)]
        FI[Query Interface 40]
        
        FE --> ND
        ND --> MG
        MG --> DB
        ND <--> FI
    end
    PLD[Property-level Data] --> FE
    Users --> FI
    FI --> DB

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(56) **References Cited**

U.S. PATENT DOCUMENTS

2016/0048934	A1 *	2/2016	Gross	G06Q 30/0643
				705/313
2016/0292800	A1 *	10/2016	Smith	G06Q 50/16
2018/0373734	A1 *	12/2018	Sargent	G06K 9/6232
2020/0012667	A1 *	1/2020	Hill	G06Q 40/10

* cited by examiner

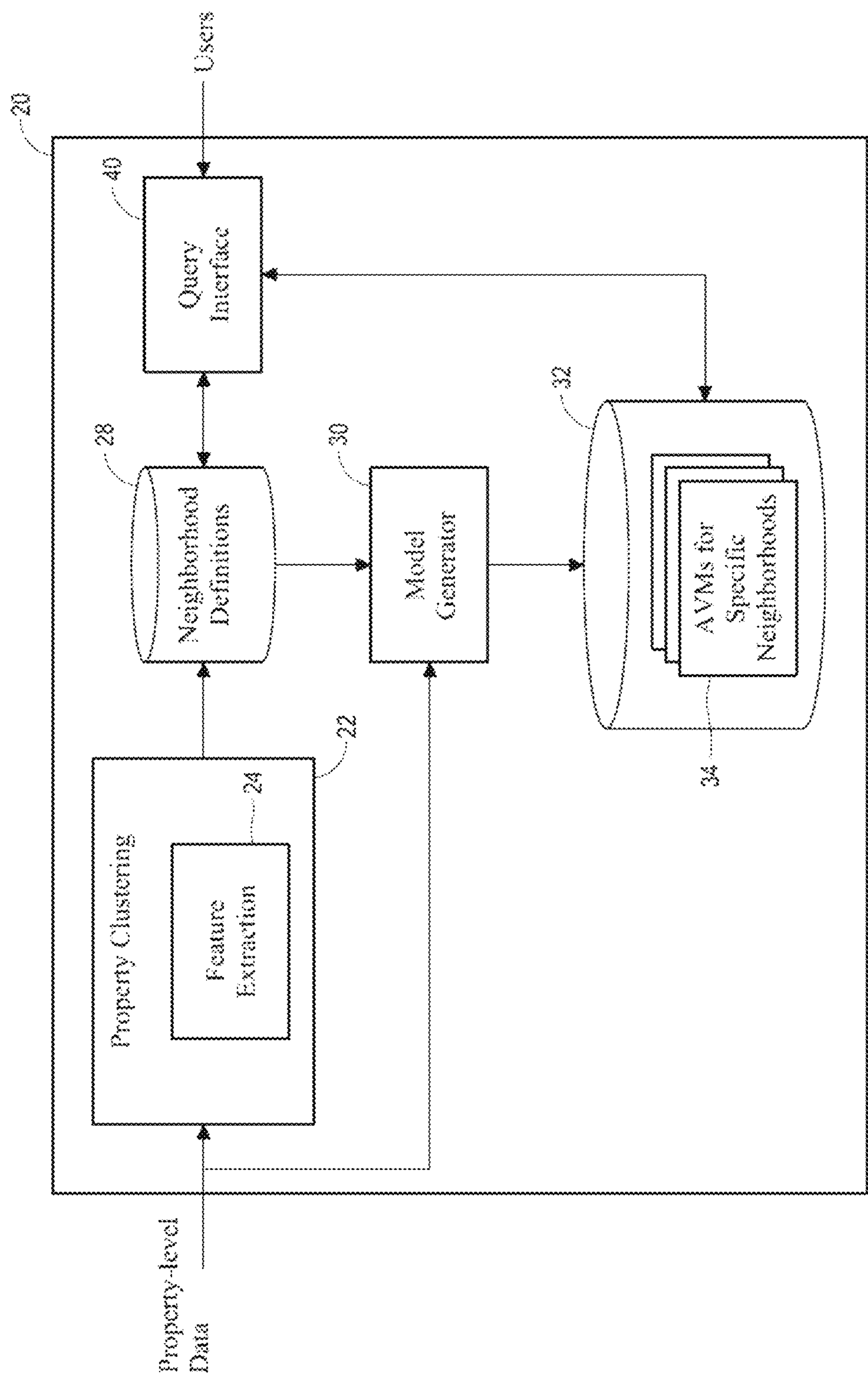


Fig. 1

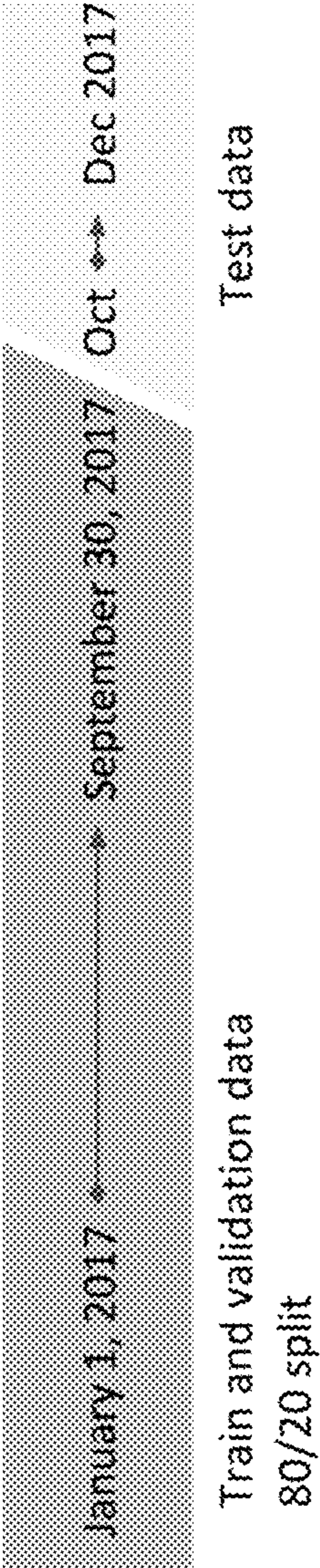


Fig. 2

Neighborhood creation method	Train coverage %	Test coverage %
Affinity Propagation and Occurrences	91.59	91.06
Affinity Propagation, Laplacian, cosine similarity	92.85	92.48
Affinity Propagation, Laplacian, Euclidean distance	92.81	92.46
Affinity Propagation, FastText, cosine similarity	92.85	92.48
Spectral Clustering	92.81	92.46
ZIP Code	99.98	100
Census Tract	65.03	57.51

Fig. 3

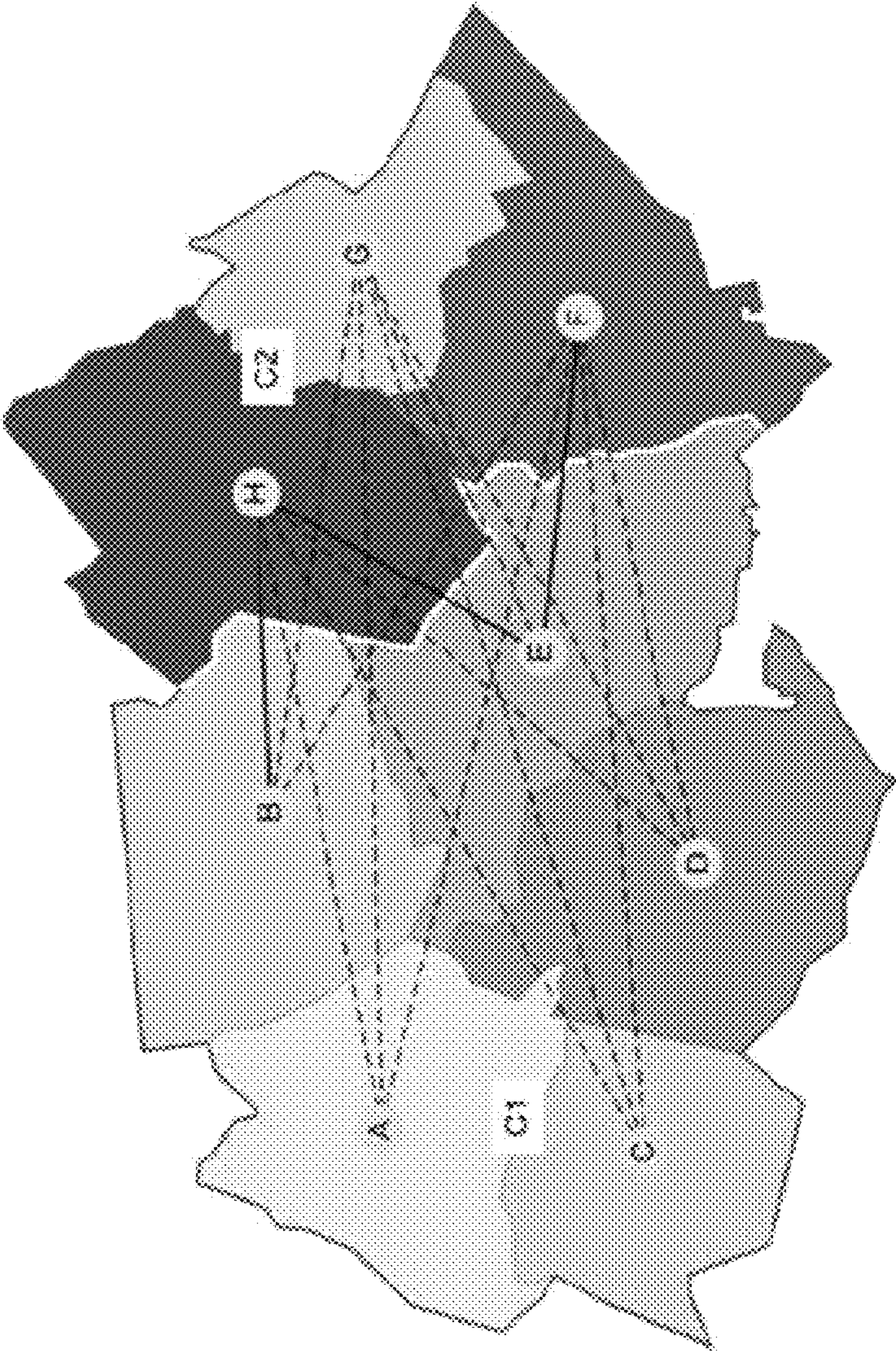


Fig. 4



Fig. 5

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COMPUTER PROCESSES FOR CLUSTERING PROPERTIES INTO NEIGHBORHOODS AND GENERATING NEIGHBORHOOD-SPECIFIC MODELS

PRIORITY CLAIM

This application claims the benefit of U.S. Provisional Appl. No. 62/815,882, filed Mar. 8, 2019, the disclosure of which is hereby incorporated by reference.

BACKGROUND

Technical Field

The present invention relates to algorithms and computer processes for grouping properties into neighborhoods.

Description of the Related Art

For various purposes it is desirable to group real estate properties into neighborhoods. For example, home price prediction algorithms, such as those implemented by automated valuation models (AVMs), commonly rely on the neighborhood of a home to predict the value of the home.

Several methods are currently used in the industry to define neighborhoods. These methods include the following: (a) grouping properties based on U.S. Census Blocks (public information created by US Census), (b) using Real Estate data or Realtor to define a commonly known neighborhoods, (c) using a group of local residents to define neighborhoods in their city, (d) using a zip codes, and dividing each zip code roughly in half, and (e) using documented “neighborhoods” defined by the City Controller. Unfortunately, these approaches often produce very inconsistent results, and results in which properties of widely ranging characteristics are assigned to the same neighborhood.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 illustrates a system for clustering properties into neighborhoods according to one embodiment.

FIG. 2 illustrates an example division of transaction data into training data, validation data, and test data.

FIG. 3 illustrates a comparison of neighborhood creation methods in terms of training coverage and test coverage.

FIG. 4 illustrates appraisal report based links created between building blocks.

FIG. 5 illustrates appraisal report based links created between residential developments.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

The present invention comprises a computer system and associated processes that use property-level data, such as MLS data, appraisal data, and mortgage records, to cluster or group “similar” properties into contiguous neighborhoods. As one example, the system may use the comparable properties identified in recent appraisal reports as one indication of properties that should be grouped into the same neighborhood (as described below). The system may assign a unique neighborhood ID to each cluster, and maintain a mapping that maps specific properties to their respective neighborhood IDs.

In some embodiments, the system also generates, for at least some of the neighborhoods, respective AVMs for

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estimating the values of properties in such neighborhoods. Each such AVM uses the information about the properties included in the respective neighborhood to make more accurate predictions of property values. The system may update the AVMs substantially in real time as new information becomes available about the properties included in the neighborhood.

FIG. 1 illustrates one embodiment of the system 20. The system includes a property clustering component 22 that receives property-level data from one or more data sources (and preferably multiple data sources), and uses this data to form property clusters that represent respective neighborhoods. The property-level data may include, for example, any one or more of the following MLS (Multiple Listing Service) data, appraisal data (e.g., appraisal reports), mortgage records, House Parcel data, and county assessor and recorder data (as aggregated and maintained, for example, by CoreLogic). The clustering component may also consider homeowner demographic data when such data is available. The system also assigns a unique neighborhood ID to each cluster.

As shown in FIG. 1, the property clustering component may include a feature extraction component 24 that determines features or characteristics of properties by analyzing the property-level data. As one example, the feature extraction component 24 may include an image analysis component that analyzes property images (e.g., as obtained from MLS records, appraisal reports, etc.) to identify the characteristics and conditions of houses. Various algorithms for detecting property characteristics and conditions from image data are known in the art and may be used for this purpose. As another example, the feature extraction component may use known text analysis algorithms to extract, from MLS listings, keywords and phrases descriptive of property characteristics and conditions.

The clustering component 22 preferably uses these extracted features, among others (such as property locations, sale prices, zip codes, etc.), to group together similar properties into neighborhoods. The clustering process may be constrained such that a contiguous boundary can be drawn around each neighborhood on a map, with each property falling within that boundary being assigned to the corresponding neighborhood. In other words, each neighborhood is preferably defined as a contiguous geographic area. Any of a variety of known clustering algorithms may be used, such as a k-means algorithm, Spectral Clustering, a hierarchical clustering algorithm, a biclustering algorithm, a self-organized map algorithm, a disjointed aggregation algorithm, or a combination thereof.

As is known in the art of clustering algorithms, different amounts of weight may be given to different property features or attributes for purposes of calculating degrees of similarity between the properties. For example, property location and listing price may be given more weight than, for example, property zip code or property condition. The weights may be selected to emphasize property attributes strongly correlated with property values, such that the neighborhood definitions are especially useful for predicting property values. In some embodiments, the clustering component gives significant weight to “comparable properties” data obtained from recent appraisal reports (e.g., appraisal reports issued in the last 3 years), such that properties identified as comparable by an appraiser will have a strong tendency to be grouped into a common neighborhood. As shown in FIG. 1, the results of the clustering process are stored in a “neighborhood definitions” database 28. This database maps specific properties (by property address,

parcel number, and/or other property identifier) to specific neighborhood IDs. The neighborhood definitions database **28** may also store other neighborhood data, such as boundary definitions (e.g., in terms of GPS coordinates), average home price, average house size, etc.

Optionally, the clustering component **22** can group together similar properties into neighborhoods by implementing some or all of the operations performed by a geocoding engine of the type described in U.S. patent application Ser. No. 14/713,663, entitled “SYSTEM AND METHOD FOR LINKING DATA RECORDS FOR PARCELS” and filed on May 15, 2015, which is hereby incorporated by reference herein in its entirety. For example, the clustering component **22** can use multiple data sources, optionally including the extracted features described above, to convert physical addresses or locations into precise geographic coordinates to identify similar properties and/or properties that fall within a contiguous boundary.

As further shown in FIG. 1, the system **20** also preferably includes a model generator **30** that generates neighborhood-specific AVMs for some or all of the defined neighborhoods. As is known in the art, the model generator **30** may use a machine learning process, such as a neural network, a support vector machine, a Bayesian network, or a Decision Tree algorithm, to determine correlations between property attributes and property values in the respective neighborhood. Each neighborhood-specific AVM may be based primarily or exclusively on the property data of the properties within the respective neighborhood. As a result, the neighborhood-specific AVMs tend to be more accurate at estimating property values than conventional AVMs. As is known in the art of machine learning and computer models, each AVM preferably includes a set of weights that specify amounts of weight to give to particular property attributes in calculating an estimated property value. In some embodiments **30**, the model generator updates the neighborhood-specific AVMs substantially in real time (e.g., on an hourly or daily basis) as new data becomes available on properties within the respective neighborhoods.

As further shown in FIG. 1, the system may also include a query interface **20** that enables users of the system to retrieve information about specific properties and/or neighborhoods. For example, in one embodiment a user, via the query interface, can enter and submit a property address (or other property identifier), and the system responds by returning some or all of the following: (1) the unique neighborhood ID of the corresponding neighborhood, (2) a map showing the boundary of the neighborhood, (3) data regarding characteristics of the neighborhood, and (4) the estimated value of the property, as calculated using the AVM for the respective property. The system may also generate an interactive map that enables the user to interactively explore the property value estimates (as calculated with the neighborhood-specific AVM) of other properties in the neighborhood.

The system **20**, including its illustrated components **22-40**, may be implemented by a computer system programmed with executable program modules stored on one or more computer-readable media (hard disk drives, solid state memory devices, etc.). The system’s components may be distributed or replicated across multiple physical servers or other computing devices of the computer system, which may or may not be co-located. Each such server typically includes one or more hardware processors that execute program instructions, a solid state memory, a network interface, and various other hardware components. The computer system may, in some embodiments, be a cloud computing

system. The functionality of the components of the system **20** may be distributed among software components differently than shown in FIG. 1, and some components and functions may be omitted in some embodiments. The data repositories **28**, **32** shown in FIG. 1 may include databases, flat file systems, and/or other types of data storage systems, and may use hard disk drives, solid state memories, and/or other types of non-transitory computer storage devices.

Use of Appraisal data

One source of data that is highly useful for auto-defining neighborhoods is appraisal data obtained from property appraisal reports. A typical appraisal report includes information about comparables. The comparables selected by the appraiser are not necessarily the most similar properties to the subject property. The appraiser typically selects comparables by selecting similar transactions which took place recently and are in the same market as the subject property. By analyzing a sufficient number of appraisals, the system can learn where the market is bounded, even though the system may not know what factors define the market (such as school district, local attraction, highway access, micro weather, scenic views, etc.).

The “market” the appraiser operates within is not equivalent to the neighborhoods defined by the system. This is largely because appraisers follow a practice of not selecting comparables further than 3 miles from the subject property. When not enough comparables can be selected, the appraisers tend to extend the search to older transactions as opposed to selecting more distant properties. To account for this common appraiser behavior, the clustering component **22** may exclude from consideration “outlier” comparables that are based on relatively old transactions, such as transactions more than 3 years before the date of the appraisal report.

The process of determining whether to connect (local) markets into a neighborhood is referred to herein as “aggregation.” When subject and comparable properties are geocoded, they become points on a map with no area. But because we can safely assume adjacent properties are likely located in the same neighborhoods, we extend the point to a tile, which has area. We call this tile a unit of aggregation. Examples of units of aggregation are a hexagon grid system, census tract, census block, and builder defined division. Once a unit of aggregation is selected, relationships between property pairs can be calculated using various measurements of similarity or “distance.” Once property similarity/distance values are calculated, one or more clustering algorithms are used to perform aggregation.

Subject-comp pairs in appraisal reports can be treated as explicit expert endorsements of two properties’ similarity. Given this, we can use these appraisals as input to data-based algorithms that seek to model and group properties based off their similarity. In one embodiment, the clustering component uses subject-comp occurrences as simple counts connecting our units of aggregation. Alternatively, more complex representations of the connections can be used using graph Spectral Clustering, which is a known connectivity-based clustering method. Regardless of the particular clustering algorithm used, the algorithm preferably treats the count of how many times two properties have been designated as comparable in appraisal reports as a measure of the similarity or relationship between these properties. The clustering algorithm may optionally give more weight to more recent appraisal reports, such as by applying an age-based decay factor during the counting process.

The primary unit of aggregation that we have explored is Uber’s Hexagonal Hierarchical Spatial Index. Hexagons are equidistant from each of their neighbors as opposed to

square or triangular grids, helping with spatial analysis and grouping. The hexagonal grid system also produces regular, non-subjective base units unlike human defined polygons. These basic units of aggregation can form the building blocks of our neighborhoods. Geographic units of aggregation are more easily defined, stable, and readily available than individual property boundaries making them preferable for use when defining neighborhoods.

Two primary categories of features affect a property's value: location attributes and physical attributes. If a region represents a neighborhood well, we will expect a good price model to depend on the physical attributes. On the contrary, if a region includes multiple neighborhoods, we will not expect a high accuracy price model to be built upon the physical characteristics only. Thus, the system evaluates a region as a candidate neighborhood by determining how well price correlates with physical attributes.

Connecting/Aggregating Neighbor Blocks Based on Appraisal Reports

In some embodiments of the system, the concept of using appraisal reports to measure the similarity between properties is extended to "neighbor blocks." The following are examples of neighbor blocks (also referred to herein simply as "blocks": (1) a multi-residence building structure or development, such as an apartment building, a condo building, or a gated community; (2) a census block or tract, or (3) a builder's subdivision. Any one or more types of neighbor blocks may be used by the property clustering component 22. As described below, a neighbor block may also be formed by merging two neighbor blocks that neighbor each other.

For example, if a condo in a condo development (one type of neighbor block) is designated as a comparable to another condo in another condo development, a link may be created between these two condo developments; this link increases the likelihood that the clustering algorithm will group these two condo developments together into a common neighborhood.

The following is one example of an algorithm that may be used by the property clustering component 22 to connect and aggregate neighbor blocks and form neighborhoods based on appraisal reports:

(1) For each comparable property pair in each appraisal report, connect (form a link between) the two neighbor blocks in which the two properties reside. This may be repeated for each of multiple types of neighbor blocks. FIGS. 4 and 5 show examples of links formed between neighbor blocks based on appraisal reports. More specifically, FIG. 4 shows links between building blocks, and FIG. 5 shows links between blocks that correspond to residential developments.

(2) Aggregate/combine the links between each pair of connected blocks, and generate a normalized score representing a degree of connectivity or similarity between the two blocks. This results in a connectivity matrix. Normalization removes the effects of block size (number of properties) may be performed by, for example, dividing the number of linkages over the total number of properties in the blocks.

(3) Apply a clustering algorithm to the connectivity matrix to identify neighboring blocks. Examples of clustering algorithms include, but are not limited to, Regionalization, Edge-Ratio Network, Maximal Entropy, and Graph Theory. Two blocks are neighbors if there are no properties between them. A clustering algorithm can, for example, use geospatial operations to determine whether two blocks are neighbors by growing/expanding each block's geographic boundary; if they touch each other without touching other

blocks first, these blocks may be deemed neighbors. The clustering algorithm may group blocks based solely on the strengths of the linkages between them (as represented, e.g., by the normalized scores), or may also consider any one or more other the other similarity factors/metrics described herein (e.g., similarity based on features extracted from property images, etc.).

(4) Merge neighboring blocks to create a new block, and form new linkages and a new connectivity matrix using the newly formed block(s).

(5) Repeat steps (3) and (4) iteratively until stopping criteria is met. The stopping criteria seeks to stop the process at a point at which neighborhoods are optimally formed. The stopping criteria may be based on information entropy, modularity, and/or other factors. As one example, a stopping criterion based on information entropy may compare the strength of the linkages within a block (which may have been formed from a block merge) to the strengths of the linkages between that block and other blocks. If the internal linkage strength is higher, the neighbor block may be considered optimal in size, and may be treated/defined as a neighborhood.

Generation of Neighborhood-Specific Price Models

Two price models that can be used to estimate property values are linear regression and XGBoost. Linear regression is a de-facto model and is widely used due to its simplicity. XGBoost is an open-source software library that provides a gradient boosting framework for various programming languages. XGBoost is a newer than linear regression, and can incorporate various additional data. Other types of gradient boosting algorithms and software packages can be used in place of XGBoost.

The system preferably uses real transaction data and the above price models to predict prices. The metrics we calculate for these price predictions are: Percent Predicted Error with 10% (PPE10), for which higher values are better, and Median Absolute Percentage Error (MAPE), for which lower values are better.

In one embodiment, the clustering component 22 runs linear and XGBoost models for each neighborhood and measures their accuracy, allowing the system to objectively assess their quality.

Evaluation and Model Results

For this study, we have chosen Los Angeles, Orange, and San Diego counties in California as well as Shelby county in Tennessee to evaluate a variety of locations. As for the historical transactions, we chose 1 year of data from four counties to evaluate our neighborhoods. 2017 was the most recent year with complete data so it was selected. This data was divided into training, validation and test data. The first 9 months of the year were used as training and validation data with an 80/20% random split. The final 3 months of the year were held out as test data to model current/future predictions. This is illustrated by FIG. 2.

The first evaluation metric we consider is coverage. We will use the numbers from Orange County to guide our discussion as seen in Table 1. Traditional ZIP codes provide complete geographic coverage but our evaluation models require a minimum amount of data to train. Additionally, linear models cannot model transactions with missing characteristics so their data coverage numbers are lower as a result. A transaction is considered "covered" if it is contained in a neighborhood and there are enough transactions to build a model for that neighborhood.

ZIP code data coverage:

XGBoost models

99.98% training data coverage

100% test data coverage

Linear models

85.12% training data coverage

77.41% test data coverage

To be viable, our neighborhoods should also provide reasonably high coverage. As seen from FIG. 3, XGBoost models on Orange County covered over 91% of the training across our methods and Table 1 shows our linear regression provided coverage with 5% of the ZIP codes.

There are two causes for the lower coverage numbers for our method. The first is that when there are insufficient historical appraisals for a tile our method does not group it into a neighborhood. This is desirable because it excludes non-residential tiles from the neighborhoods. The results in less than 100% geographic coverage so some percent of the evaluation data will not fall into a neighborhood. The second cause is the same as we saw with ZIP codes: some of the neighborhoods have insufficient evaluate data to train a model. These combine to give the coverage numbers seen in FIG. 3.

The next metrics to explore are the model fidelity metrics. We are already seeing positive results that improve over the baseline ZIP code groupings. As seen in Table 1, our neighborhoods are showing 5% improvement in PPE10 on both the validation and test data when compared to ZIP codes as well as a 1% improvement in MAPE. The PPE10 improvements are particularly exciting because this is a key measure of accuracy in the property valuation domain and shows that our neighborhoods already provide a significant advantage.

TABLE 1

Method	Train Coverage	Test Coverage	PPE10 Train	PPE10 Valid	PPE10 Test	MAPE Train	MAPE Valid	MAPE Test
Affinity propagation and occurrence - XGBoost	91.59	91.06	84.7%	56.4%	49.6%	2.7%	8.3%	10.1%
Linear regression	80.83	72.82	56.5%	53.6%	50.8%	8.3%	9.1%	9.8%
ZIP - XGBoost	99.98	100	62.8%	51.1%	47.7%	6.9%	9.7%	11.8%
Linear regression	85.12	77.41	48.9%	46.8%	46.4%	10.3%	11.0%	10.9%

All of the methods and tasks described herein may be performed and fully automated by a computer system. The computer system may, in some cases, include multiple distinct computers or computing devices (e.g., physical servers, workstations, storage arrays, etc.) that communicate and interoperate over a network to perform the described functions. Each such computing device typically includes a processor (or multiple processors) that executes program instructions or modules stored in a memory or other non-transitory computer-readable storage medium. The various functions disclosed herein may be embodied in such program instructions, although some or all of the disclosed functions may alternatively be implemented in application-specific circuitry (e.g., ASICs or FPGAs) of the computer system. Where the computer system includes multiple computing devices, these devices may, but need not, be co-located. The results of the disclosed methods and tasks may

be persistently stored by transforming physical storage devices, such as solid state memory chips and/or magnetic disks, into a different state.

Although this invention has been disclosed in the context of certain preferred embodiments and examples, it will be understood by those skilled in the art that the present invention extends beyond the specifically disclosed embodiments to other alternative embodiments and/or uses of the invention and obvious modifications and equivalents thereof. Thus, it is intended that the scope of the present invention herein disclosed should not be limited by the particular disclosed embodiments described above, but should be determined only by a fair reading of the claims that follow.

What is claimed is:

1. A computing system comprising one or more computing devices programmed with executable code to implement at least:

a feature extraction component that extracts property features from property-level data associated with real estate properties; and

a clustering component that uses at least the extracted property features to group properties into neighborhoods based on property similarity such that each neighborhood is a contiguous geographic neighborhood, wherein the clustering component gives different amounts of weight to different property features for purposes of measuring property similarity;

wherein the property-level data comprises appraisal reports that identify specific properties as comparable, and the clustering component uses the identifications of comparable properties as a factor in measuring property similarity.

2. The computing system of claim 1, wherein the feature extraction component analyzes property images to extract features associated with characteristics and conditions of houses, and the clustering component uses said features associated with characteristics and conditions of houses to measure property similarity.

3. The computing system of claim 1, wherein the clustering component generates, for a property pair composed of a first property and a second property, an occurrence count value representing a number of times the first and second properties have been designated as comparable in the appraisal reports, and further uses said count value as a factor for measuring the property similarity of the first and second properties.

4. The computing system of claim 1, wherein the clustering component is configured to measure similarity between two neighbor blocks, each of which comprises a

plurality of properties, based at least in part on appraisal report based links between the properties in the respective neighbor blocks.

5 **5.** The computing system of claim **4**, wherein the clustering component is configured to merge neighbor blocks based at least in part on measures of similarity between the neighbor blocks.

6. The computing system of claim **4**, wherein the model generation component is configured to use a machine learning process to determine, for a specific neighborhood, correlations between property attributes and property values in the specific neighborhood, and is configured to generate weight values specifying amounts of weight to give to specific property attributes in estimating property values in the specific neighborhood.

7. The computing system of claim **1**, wherein the computing system is further programmed with the executable code to implement a model generation component that generates, for individual neighborhoods formed by the clustering component, neighborhood-specific Automated Valuation Models (AVMs), each neighborhood-specific AVM configured to estimate values of properties in its respective neighborhood.

8. A computing system comprising one or more computing devices programmed with executable program code to implement a process that comprises:

identifying a plurality of neighbor blocks within a geographic region, each neighbor block comprising a plurality of residential properties;

creating links between pairs of the neighbor blocks based on appraisal reports, each link based on a property in one of the neighbor blocks being identified in an appraisal report as comparable to another property in another neighbor block; and

applying a clustering algorithm to the neighbor blocks to form at least one contiguous geographic neighborhood, wherein the clustering algorithm groups together neighbor blocks based at least partly on numbers of said links formed between the neighbor blocks.

9. The computing system of claim **8**, wherein the clustering algorithm additionally takes into consideration data extracted from property images.

10. The computing system of claim **8**, wherein the process comprises generating a normalized score representing a strength of the links created between two of the neighbor blocks, wherein the clustering algorithm uses at least the normalized score to determine whether to group together the two neighbor blocks into a common neighborhood.

11. The computing system of claim **8**, wherein applying the clustering algorithm comprises determining whether to merge together two neighbor blocks to form a new neighbor block.

12. The computing system of claim **11**, further comprising, after a neighbor block merge operation, reapplying the clustering algorithm, and determining whether to merge any additional neighbor blocks.

13. The computing system of claim **12**, wherein determining whether to merge any additional neighbor blocks comprises, for a selected neighbor block formed by merging two or more other neighbor blocks, determining whether a strength of the links within the selected neighbor block

exceeds a strength of links between the selected neighbor block and other neighbor blocks.

14. The computing system of claim **8**, wherein the process further comprises generating, for a neighborhood formed by application of the clustering algorithm, a neighborhood-specific model configured to estimate values of properties in the neighborhood.

15. The computing system of claim **14**, wherein generating the neighborhood-specific model comprises using a machine learning process to determine, for the neighborhood, correlations between property attributes and property values in the neighborhood, and generating weight values specifying amounts of weight to give to specific property attributes in estimating property values in the neighborhood.

16. A process performed by one or more computing devices under control of executable program instructions, the process comprising:

extracting property features from property-level data associated with real estate properties, said property-level data comprising appraisal reports that identify specific properties as comparable; and

grouping properties into neighborhoods based at least partly on property similarity such that each neighborhood is a contiguous geographic neighborhood, wherein grouping the properties into neighborhoods comprises using the identifications of comparable properties in the appraisal reports as a factor in measuring property similarity, and additionally comprises giving different amounts of weight to different ones of said extracted property features.

17. The process of claim **16**, wherein extracting said property features comprises extracting, from property images, features associated with characteristics and conditions of houses, and wherein grouping the properties into neighborhoods comprises using said features associated with characteristics and conditions of houses as a factor for measuring property similarity.

18. The process of claim **16**, wherein grouping the properties into neighborhoods comprises, for a property pair composed of a first property and a second property, generating an occurrence count value representing a number of times the first and second properties have been designated as comparable in the appraisal reports, and further comprises using said occurrence count value as a factor for measuring the property similarity of the first and second properties.

19. The process of claim **16**, wherein grouping the properties into neighborhoods comprises measuring similarity between two neighbor blocks, each of which comprises a plurality of properties, based at least in part on appraisal report based links between the properties in the respective neighbor blocks.

20. The process of claim **19**, wherein grouping the properties into neighborhoods further comprises merging neighbor blocks based at least in part on measures of similarity between the neighbor blocks.

21. The process of claim **16**, further comprising generating, for a first neighborhood of said neighborhoods, a neighborhood-specific Automated Valuation Model (AVM) configured to estimate values of properties in the first neighborhood.